Lecture 3.1: Signaling System No. 7

Content

- SS7 system principle
- Message transfer part (MTP) principle
- ISDN user part (ISUP) message and application

Signaling System No.7 (SS7)

- Signaling is used for connection control, work coordination and "session" between communication devices
- SS7 is formulated by CCITT (Consultative Committee for International Telephony and Telegraphy)and is widely applied in switched circuit networks (e.g. PSTN).

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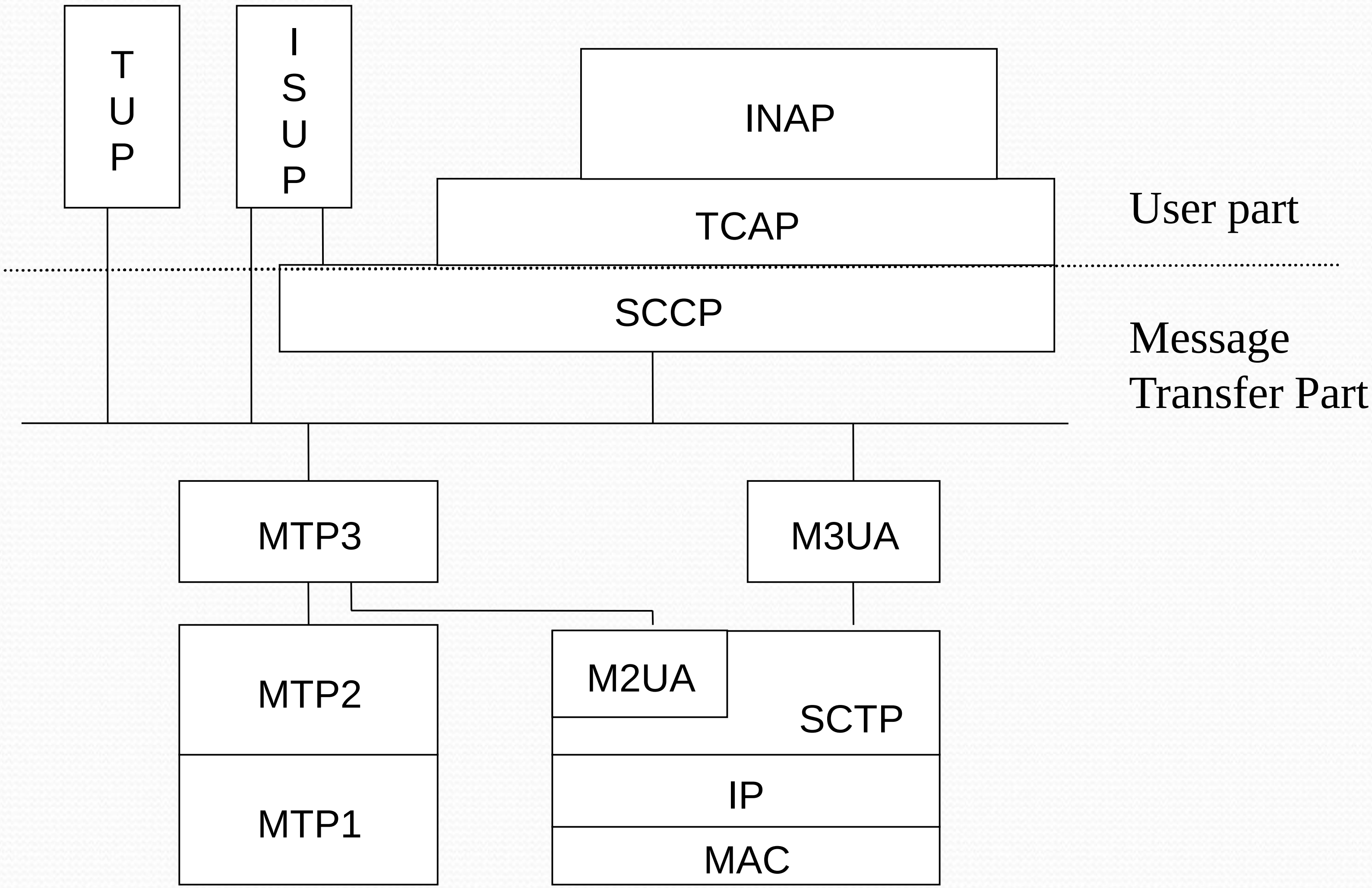
SS7 Functional Structure



SS7 Functional Structure

- SS7 has two parts:
  - User Part (UP)
  - Message Transfer Part (MTP).
- MTP : enables reliable transfer of signaling messages between user functions.
  - SS7 messages can be transmitted over
    - narrowband TDM transmission network (i.e. MTP),
    - over broadband IP network (the protocol used for transmitting SS7 over IP network is SIGTRAN.)
- UP :
  - the independent user part of different users
  - it is the functional entity for various call services

MTP

- Overview, MTP1, MTP2
- MTP3

MTP - Overview

- The main function
  - to provide reliable signaling message transfer in the signaling network
- Feature : has measures to avoid or reduce
  - message loss
  - repetition
  - loss of sequence
- MTP consists of three function levels:
  - signaling data link (MTP1)
  - signaling link function (MTP2)
  - signaling network function (MTP3).

MTP1

- MTP1 definition:
  - Defining the physical, electric and functional characteristics of signaling data link, as well as access methods.
  - Equivalent to the physical layer of OSI seven-layer protocol structure, and used to generate and receive the signals over physical channels
- Signaling Data Link :
  - A bidirectional transmission channel for a signaling.
  - Composed of two-way data channels.
  - The standard bit-rate is 64kbit/s
    - which also can be applied to the transmission link with lower rate (such as 4.8kbit/s) or that with higher rate (2048kbit/s)

MTP2

- MTP2 definition:
  - Having a Signaling link function.
  - It is used to transmit signaling to data link, and cooperates with MTP1 to ensure reliable signaling link between two directly connected signaling points.
- Signaling link function can be sub-divided into
  - signaling unit delimitation
  - signaling unit location
  - error detection
  - error correction,
  - initial location
  - processor failure
  - level-two flow control
  - and signaling link error rate monitoring functions.

MTP

- Overview, MTP1, MTP2
- MTP3

MTP3

- Definition :
  - Network layer
  - It implements the functions of the third layer of OSI
  - used to transmit management messages to ensure reliable transmission of signaling messages in case of faults in signaling link or signaling transfer point

Signaling Message Processing

- Message routing function:
  - To determine the outgoing signaling link for transmitting messages to destination point
- Message identification function:
  - To check whether the destination signaling point in received message is a local office signaling point
- Message allocation function:
  - To distribute received messages (sent to the point) to corresponding UPs

Signaling Network Management

- Signaling service management:
  - To transfer signaling service from one link or route to one or multiple different links or routes, reset MTP in a signaling point or slow down the traffic in case of congestion
- Signaling link management:
  - To recover faulty signaling link, activate idle (non-arranged) links, and deactivate arranged signaling links.
- Signaling route management:
  - To distribute the messages related to signaling network status, block or unblock signaling routes

Message Format

- MTP has three basic signaling unit formats:
  - Message Signaling Unit (MSU)
    - To transmit messages of respective UPs, signaling network management messages as well as signaling network test and maintenance messages.)
  - Link Status Signaling Unit (LSSU)
    - LSSU provides the link status information to perform the connection and recovery of signaling links
  - Fill-In Signaling Unit (FISU)
    - FISU is used to maintain normal operation of the signaling link and implement fill-in function when no MSU or LSSU is transmitted over the signaling link.

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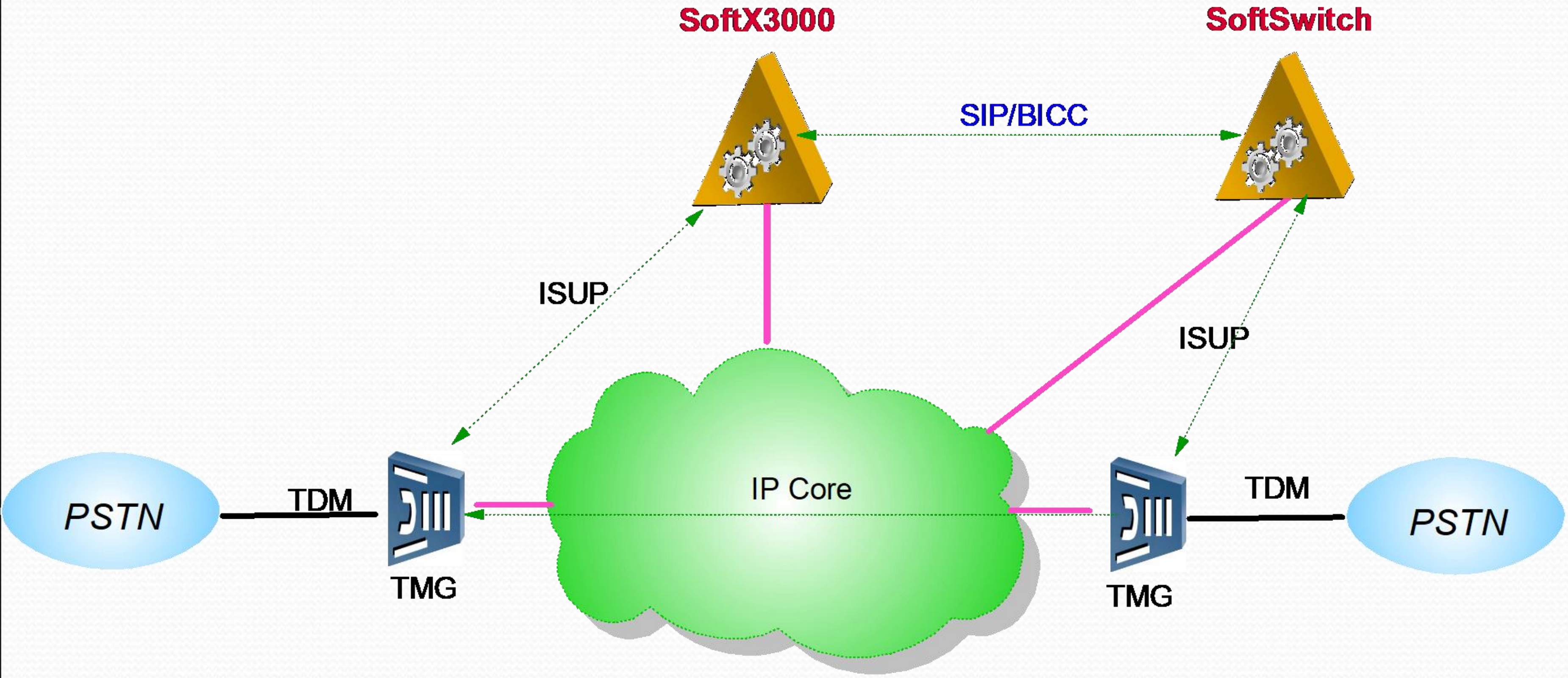
ISUP Signaling

- Overview
- Call process

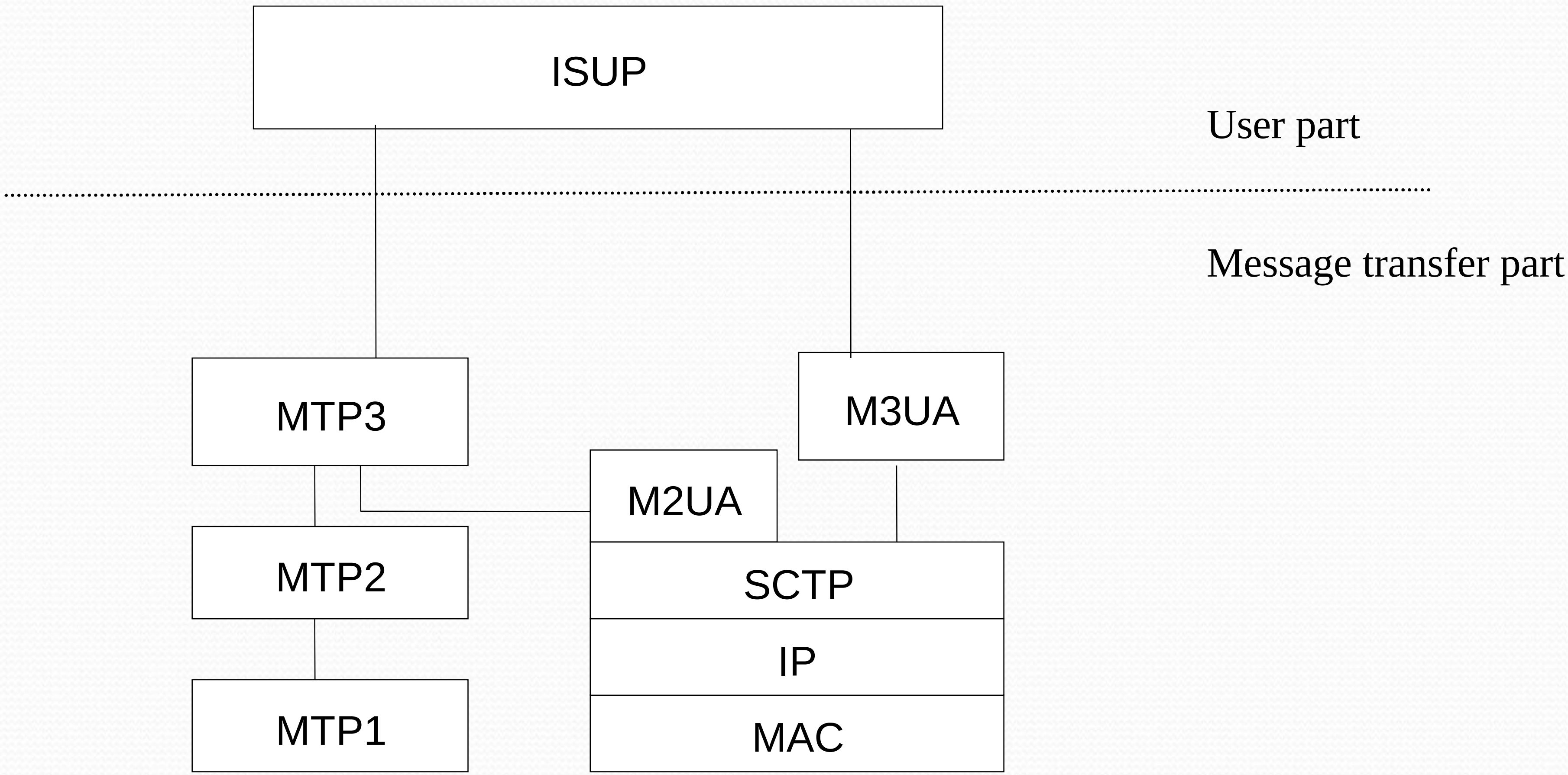
ISUP - Overview

- ISUP is one of UPs of Common Channel Signaling System No.7.
- It provides necessary signal functions for supporting basic bearer services and supplementary services of voice and non-voice purposes in the Integrated Service Digital Network (ISDN).

Applications in SoftX3000



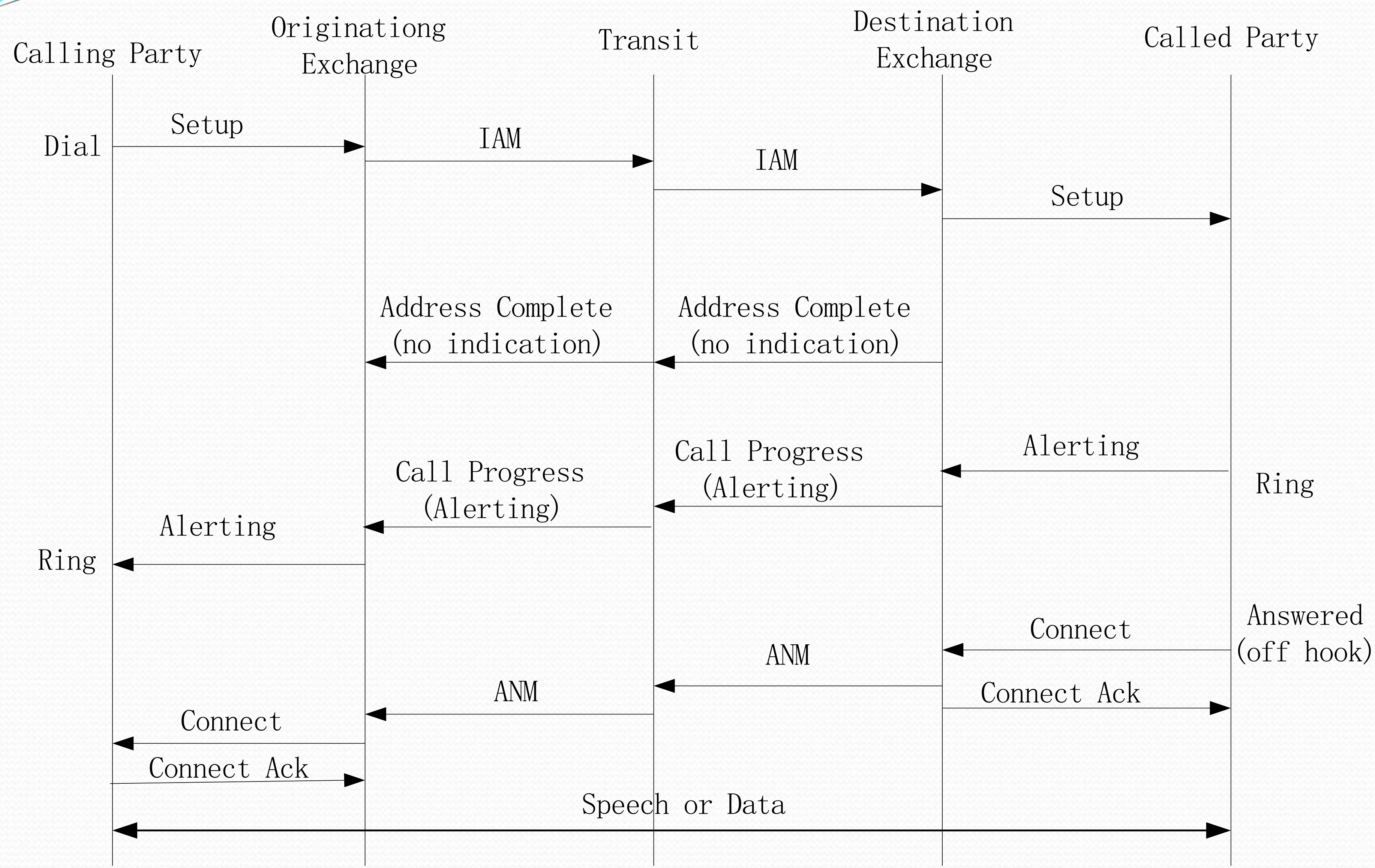
Protocol Stack



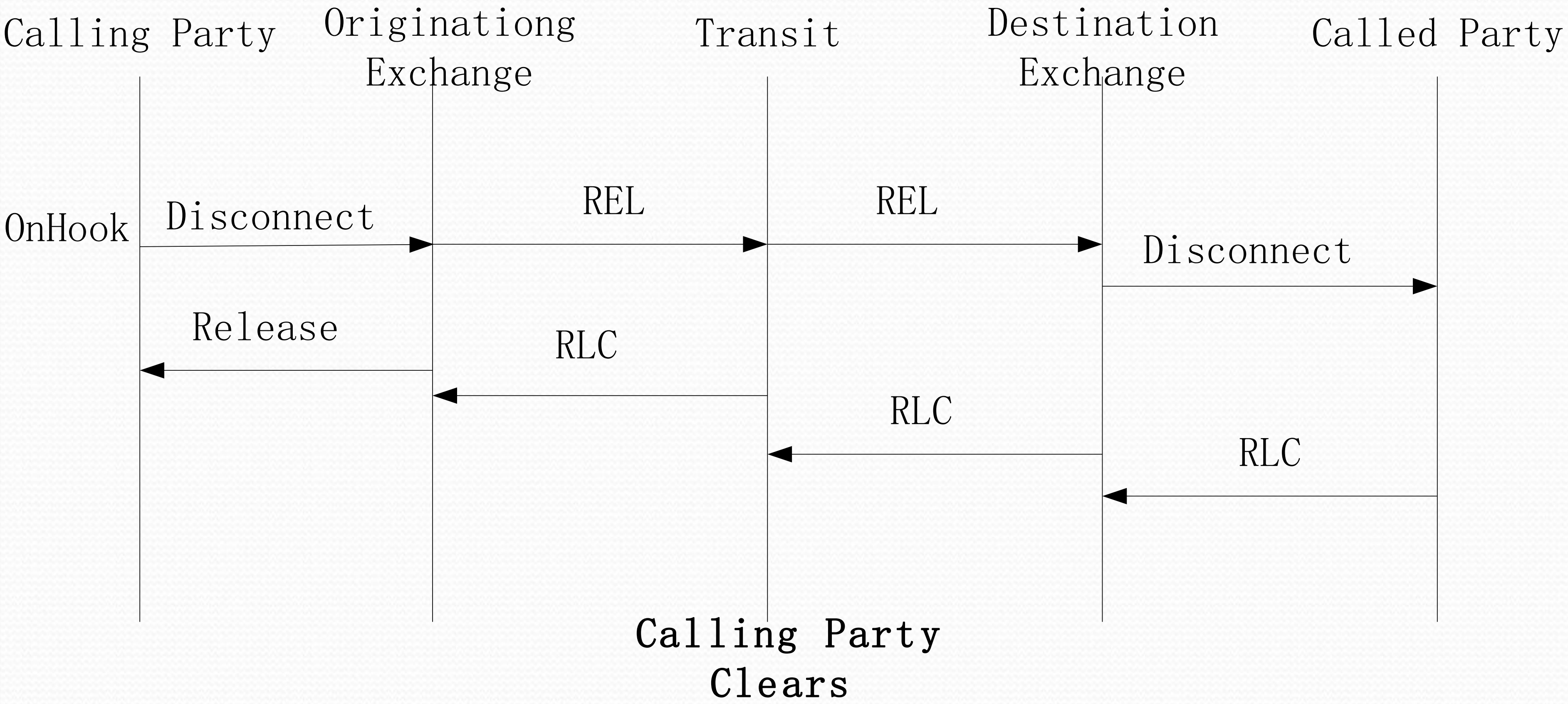
ISUP Signaling

- Overview
- Call process

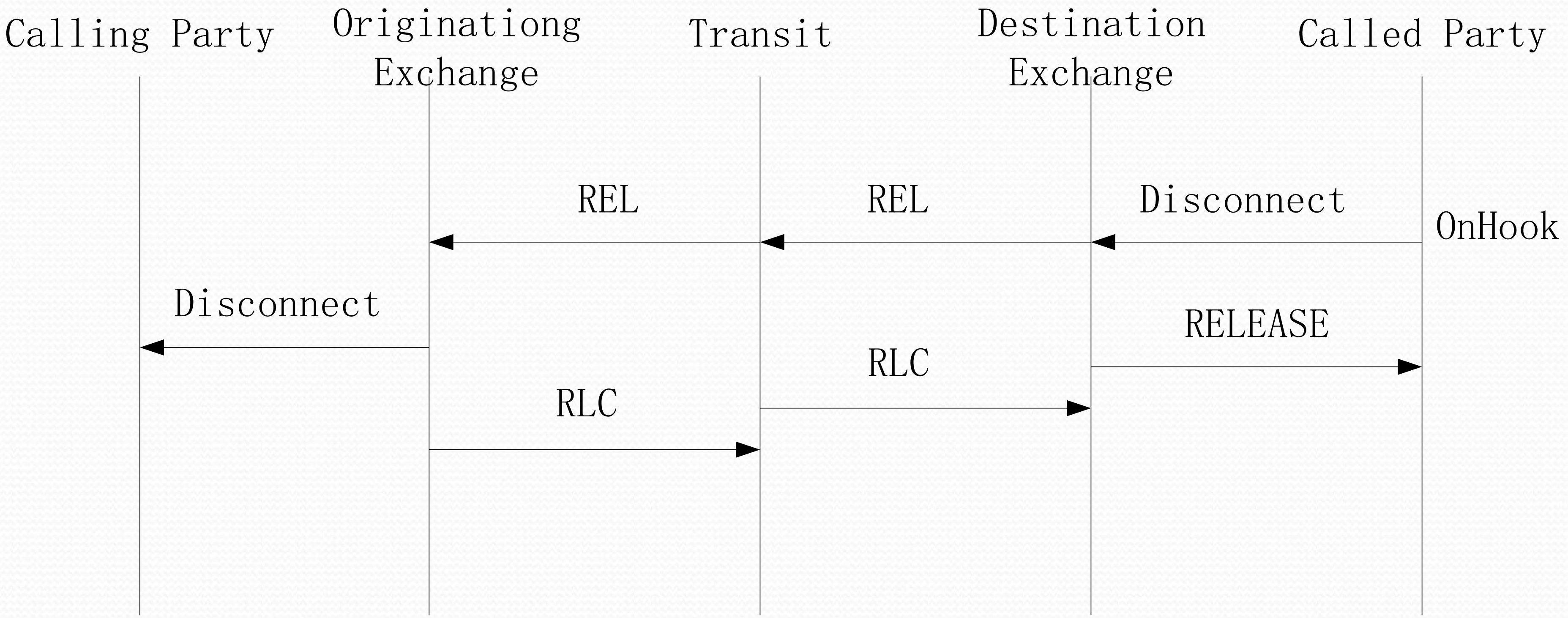
Successful Call Setup



Successful Call Release (1)



Successful Call Release (2)



Called Party Clears

Summary

- SS7 signaling includes two parts: Message Transfer Part (MTP) and User Part (UP).
- The MTP consists of MTP1, MTP2 and MTP3
- ISUP is one of UPs of SS7. It provides necessary signal functions for supporting basic bearer services and supplementary services of voice and non-voice purposes in the integrated service digital network.